

CHARLO, NB, Canada

WMO: 713150

Lat: 47.9889N

Lon: 66.3336W

Elev: 138

StdP: 14.62

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-7.6	-22.2	1.6	-10.4	-17.6	2.1	-6.5	27.3	17.3	25.0	15.5	8.1	260	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	83.3	67.7	79.5	65.9	76.3	64.3	70.4	79.5	68.3	76.4	66.5	73.3	8.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	100.2	75.2	65.3	94.1	72.5	63.6	88.5	70.4	34.4	79.5	32.7	76.8	31.2	73.4	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	20.8	18.5	-18.7	89.3	3.7	2.7	-21.3	91.2	-23.4	92.8	-25.5	94.3	-28.1	96.2
			-19.1	73.8	3.6	1.6	-21.6	74.9	-23.7	75.9	-25.7	76.7	-28.3	77.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.8	12.0	14.1	23.9	36.0	48.6	58.8	65.6	64.0	55.5	44.2	32.5	20.3
	DBStd	20.11	11.13	10.68	9.61	6.31	6.59	6.69	5.16	5.18	6.72	6.72	8.38	9.72
	HDD50	5201	1179	1005	808	421	107	6	0	0	21	204	528	922
	HDD65	9364	1644	1425	1273	869	511	208	54	80	292	645	976	1387
	CDD50	1468	0	0	0	2	63	270	484	435	188	24	2	0
	CDD65	156	0	0	0	0	2	22	73	51	8	0	0	0
	CDH74	1237	0	0	0	1	46	241	508	361	76	4	0	0
	CDH80	281	0	0	0	0	9	61	106	89	16	0	0	0
Wind	WSAvg	8.0	8.9	9.1	9.1	9.0	8.3	7.5	6.5	6.3	6.8	7.9	8.1	8.7
Precipitation	PrecAvg	43.8	3.7	2.7	3.5	3.5	3.5	3.5	3.9	4.3	3.5	3.6	3.7	4.4
	PrecMax	52.7	7.1	5.2	7.4	7.7	7.0	7.7	8.3	6.3	6.9	8.1	9.8	7.5
	PrecMin	32.2	0.5	0.4	1.5	0.9	0.6	1.4	1.5	2.4	0.7	0.9	1.5	2.4
	PrecStd	6.1	1.4	1.2	1.6	1.7	1.7	1.4	1.7	1.1	1.7	1.8	1.7	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.6	42.7	52.4	63.9	80.3	86.7	87.5	87.7	81.9	71.0	58.9	42.9
		MCWB	37.4	39.5	44.8	48.9	59.6	67.4	69.7	70.1	68.1	59.0	51.8	41.0
	2%	DB	35.2	36.5	42.6	55.0	71.8	80.6	83.0	82.0	75.1	62.7	50.6	38.9
		MCWB	33.4	34.4	36.1	44.0	56.7	64.7	68.4	68.3	64.5	55.2	46.9	36.7
	5%	DB	32.0	33.0	38.9	49.5	65.8	75.7	79.3	77.7	70.7	57.5	46.7	35.9
		MCWB	30.7	30.9	33.9	41.1	53.8	62.2	66.5	65.8	62.0	51.6	43.5	33.9
	10%	DB	28.6	30.1	35.9	45.4	60.8	71.1	76.4	74.2	66.9	54.0	43.3	33.6
		MCWB	27.2	28.3	32.3	38.5	50.8	59.7	64.8	64.0	59.7	49.6	40.5	32.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.9	39.5	44.8	51.3	63.7	70.9	73.1	72.5	70.0	61.5	55.2	41.4
		MCDB	39.3	42.3	49.3	59.8	74.3	82.3	82.5	83.2	78.4	67.8	58.8	42.5
	2%	WB	33.6	34.6	37.5	45.0	58.8	67.2	70.6	69.9	66.3	56.2	48.0	37.1
		MCDB	35.1	36.2	41.7	53.0	67.7	76.3	79.4	79.1	72.9	61.0	49.8	38.4
	5%	WB	30.6	31.3	34.8	41.9	55.3	64.5	68.6	67.7	63.3	52.8	43.9	34.3
		MCDB	31.8	32.7	37.8	48.0	63.6	71.6	76.4	75.3	69.1	56.5	45.9	35.4
	10%	WB	27.2	28.3	32.8	39.4	52.1	61.9	66.7	65.8	60.7	50.1	40.8	32.2
		MCDB	28.5	30.0	35.4	43.9	58.8	68.8	73.6	71.8	65.6	53.1	43.1	33.4
Mean Daily Temperature Range	5% DB	MDBR	15.0	16.1	15.5	14.1	18.0	18.0	17.4	17.8	17.6	13.7	11.3	12.3
		MCDBR	16.5	15.5	17.8	22.6	28.2	26.7	23.0	23.6	22.8	20.2	15.6	11.7
		MCWBR	16.1	13.6	13.0	13.9	15.6	13.8	11.6	12.2	14.1	13.9	13.3	11.1
	5% WB	MCDBR	16.5	14.2	15.7	20.1	24.3	21.1	19.2	20.0	19.8	18.1	14.7	11.1
		MCWBR	16.2	13.2	12.3	13.5	15.2	12.5	10.8	11.3	13.5	13.7	13.4	11.1
Clear-Sky Solar Irradiance	taub	0.272	0.280	0.318	0.419	0.366	0.384	0.425	0.393	0.366	0.335	0.299	0.274	
	taud	2.328	2.287	2.265	2.091	2.378	2.348	2.268	2.357	2.413	2.492	2.508	2.404	
	Ebn at Noon	256	280	285	263	285	279	265	269	266	258	247	242	
	Edh at Noon	23	29	36	47	37	38	41	36	31	24	19	19	
All-Sky Solar Radiation	RadAvg	411	686	1069	1394	1599	1733	1697	1478	1137	678	367	279	
	RadStd	48	79	77	123	136	128	157	94	88	59	47	28	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	+3.71	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	N/A	N/A	+2.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page